Transfusion on-call: A haematologist's cheat sheet

Crossing ABO and/or Rh groups

Red cells

Recommendations for the emergency use of group O uncrossmatched red blood cells (RBC)

- Emergency uncrossmatched group O RBC are to be used only in an emergency to save a patient's life and when there is no current valid pretransfusion specimen, not as a time-saving measure.
- Where a patient's ABO RhD blood group is unknown or has not been confirmed by current pretransfusion testing, the following applies:

  For females of childbearing potential (≤50 years) and paediatric males ≤18 years (or as per local paediatric policy)
  Issue **group O RhD negative uncrossmatched RBC** until the patient's ABO RhD blood group has been determined using a valid current specimen.

  For females >50 years and all adult males >18 years
  Issue **group O RhD positive uncrossmatched RBC** until the patient's ABO RhD blood group has been determined using a valid current specimen.
- The patient's ABO group must be determined as soon as possible to allow ABO RhD group specific RBCs to be issued.
- When four units of uncrossmatched group O RhD negative RBC have been issued to a patient of unknown RhD group, transfusion of group O RhD positive RBC should be considered irrespective of age, gender or child-bearing potential.

Source: National Statement for the Emergency Use of Group O Red Blood Cells

Red cell ABO compatibility table

Patient ABO group	Donor ABO group 1st choice	Donor ABO group 2nd choice	Never use
O	O	None	A, B, AB
A	A	O	B, AB
B	B	O	A, AB
AB	AB	A, B, O	None

Source: Component compatibility | Australian Red Cross Lifeblood

Red cell RhD compatibility table

Patient RhD group	Donor RhD group 1st choice	Donor RhD group 2nd choice
RhD negative	RhD negative	RhD positive: In an emergency, when RhD negative components are not available, transfusion should not be withheld.
RhD positive	RhD positive	RhD negative

Source: Component compatibility | Australian Red Cross Lifeblood

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Platelets

Platelet compatibility table

Patient ABO group	Donor ABO group 1st choice	Donor ABO group 2nd choice	Donor ABO group 3rd choice
Unknown (emergency issue)	O or A*	B*	
O	O	A	B
A	A	B*, O*	
B	B	A*, O*	
AB	AB (not routinely available)	A*, B*	O*

*Low-titre anti-A/B apheresis or pooled platelets pose lower risk of haemolysis when transfusing across ABO groups. Pooled and apheresis platelets are resuspended in platelet additive solution (PAS). This reduces the residual plasma content and reduces the risk of:

- allergic reactions
- haemolytic transfusion reactions when crossing ABO blood groups.

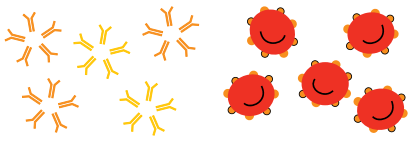
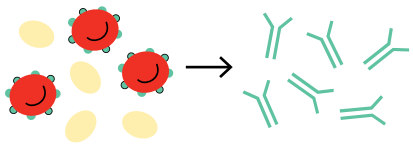
Source: Component compatibility | Australian Red Cross Lifeblood

Apheresis vs pooled platelets

	Apheresis	Pooled
Volume (mL)	214 ± 11 (100-400)	259 ± 11 (>160)
Platelet count (x10^9/L)	284 ± 36 (>200-<400)	257 ± 43 (>200)
Residual plasma (% total volume)	Approx. 40%	Approx. 30%

Mean ± standard deviation (Specification requirements)

Types of incompatible platelet transfusions

Donor plasma is incompatible with recipient ABO red cell antigens	Donor ABO platelet antigens are incompatible with recipient plasma	Donor residual red cells in platelet transfusion are Rh incompatible
Risk: ABO haemolytic transfusion reaction	**Risk:** Poor platelet increments	**Risk:** Rh alloimmunisation
Mechanism: **Donor:** Anti-A and Anti-B antibodies **Recipient:** RBCs express A and/or B antigen  **To reduce the risk:** Transfuse low titre anti-A and/or anti-B apheresis platelets	**Mechanism:** **Donor:** Platelets express A and/or B antigen **Recipient:** Anti-A and Anti-B antibodies 	**Mechanism:** **Donor:** RBCs express D antigen **Recipient:** Forms Anti-D antibody  **Note:** If RhD positive platelets are given to an RhD negative patient, who is a female of childbearing potential or requires regular transfusion, RhD Ig should be considered. 250IU RhD immunoglobulin IM provides sufficient cover for six weeks of platelet transfusions. Rophylac can be considered for IV administration.

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Plasma

Recommendations for the emergency use of a group A clinical plasma (FFP/ELP/cryoprecipitate)

- Use of group A FFP, ELP or cryoprecipitate with low anti-A/B titre is recommended in traumas and emergencies where the patient's ABO group is unknown. For neonatal patients, use group AB plasma.
- It is important that the patient's ABO group is determined as soon as possible to allow ABO identical or compatible plasma to be issued.

Notes:

- All clinical apheresis donors undergo anti A/B titre testing and associated components are labelled low titre accordingly.
- ELP is thawed FFP stored at 2-6 °C that can be transfused for up to 5 days after thawing.

Source: National Statement for the Emergency Use of Group A Clinical Plasma

Plasma compatibility table (fresh frozen plasma and cryoprecipitate)

Patient ABO group	Donor ABO group 1st choice	Donor ABO group 2nd choice	Donor ABO group 3rd choice	Donor ABO group 4th choice
O	O	A	B	AB
A	A	AB		
B	B	AB or A (low titre)		
AB	AB	A (low titre)		
Adults and paediatrics Unknown (emergency issue)	A (low titre)	AB	A	
Neonates and infants <1 yr old Unknown (emergency issue)	AB (low titre)	A (low titre)	A (titre unknown)	

Low titre = low-titre anti-A/B